```
# Install required libraries
!pip install transformers torch --quiet
```

```
# Import libraries
from transformers import pipeline
import torch

print("Libraries installed successfully!")
```

```
Libraries installed successfully!
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

tokenizer = AutoTokenizer.from_pretrained("facebook/bart-large-cnn")
model = AutoModelForSeq2SeqLM.from_pretrained("facebook/bart-large-cnn")

print("Model and tokenizer loaded successfully!")
```

```
Loading weights: 100%|| 511/511 [00:00<00:00, 617.70it/s, Materializing param=model.encoder.layers.11.self_attn_layer_norm.weight]
Model and tokenizer loaded successfully!
```

```
# Sample dataset (you can edit or replace these)

texts = [

"""Artificial Intelligence is transforming industries by enabling machines to perform tasks
that traditionally required human intelligence. These include speech recognition, decision-making,
visual perception, and language translation. AI systems are increasingly used in healthcare, finance,
and education to improve efficiency and accuracy. However, ethical concerns such as data privacy,
bias, and job displacement remain significant challenges.""",

"""Climate change is one of the most pressing global challenges today. Rising temperatures, melting
ice caps, and extreme weather events are clear indicators of environmental imbalance. Governments
and organizations worldwide are working towards reducing carbon emissions and promoting sustainable
energy sources. Individual actions, such as reducing waste and conserving energy, also play a
crucial role.""",

"""In software development, version control systems like Git help developers manage code changes
efficiently. They allow multiple contributors to work on the same project without conflicts.
GitHub provides a platform for collaboration, issue tracking, and deployment. Understanding Git
commands is essential for modern developers.""",

"""Once upon a time, a young girl discovered a hidden door in her grandmother’s house. Behind the
door was a magical world filled with glowing trees and talking animals. She embarked on an adventure
to save the kingdom from darkness. Her courage and kindness helped restore peace.""",

"""Customer: I am facing issues with my internet connection.
Support: Please restart your router and check the cables.
Customer: I tried that but it still does not work.
Support: I will register a complaint and a technician will visit your location tomorrow."""
]

print("Dataset loaded:", len(texts), "texts")
```

```
Dataset loaded: 5 texts
```

```
# Generate summaries

results = []

for i, text in enumerate(texts):
    inputs = tokenizer([text], max_length=1024, return_tensors="pt", truncation=True)

    # Generate Summary
    summary_ids = model.generate(
        inputs["input_ids"],
        num_beams=4,
        min_length=20,
        max_length=60,
        early_stopping=True
    )
```

```
    summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

    print(f"\n--- TEXT {i+1} ---")
    print("Original Text:\n", text)
    print("\nSummary:\n", summary_text)

    results.append(summary_text)
```

```
--- TEXT 1 ---
Original Text:
 Artificial Intelligence is transforming industries by enabling machines to perform tasks
that traditionally required human intelligence. These include speech recognition, decision-making,
visual perception, and language translation. AI systems are increasingly used in healthcare, finance,
and education to improve efficiency and accuracy. However, ethical concerns such as data privacy,
bias, and job displacement remain significant challenges.

Summary:
 Artificial Intelligence is transforming industries by enabling machines to perform tasks that traditionally required human inte

--- TEXT 2 ---
Original Text:
 Climate change is one of the most pressing global challenges today. Rising temperatures, melting
ice caps, and extreme weather events are clear indicators of environmental imbalance. Governments
and organizations worldwide are working towards reducing carbon emissions and promoting sustainable
energy sources. Individual actions, such as reducing waste and conserving energy, also play a
crucial role.

Summary:
 Climate change is one of the most pressing global challenges today. Governments and organizations worldwide are working towards

--- TEXT 3 ---
Original Text:
 In software development, version control systems like Git help developers manage code changes
efficiently. They allow multiple contributors to work on the same project without conflicts.
GitHub provides a platform for collaboration, issue tracking, and deployment. Understanding Git
commands is essential for modern developers.

Summary:
 GitHub provides a platform for collaboration, issue tracking, and deployment. Understanding Git commands is essential for moder

--- TEXT 4 ---
Original Text:
 Once upon a time, a young girl discovered a hidden door in her grandmother’s house. Behind the
door was a magical world filled with glowing trees and talking animals. She embarked on an adventure
to save the kingdom from darkness. Her courage and kindness helped restore peace.

Summary:
 Once upon a time, a young girl discovered a hidden door in her grandmother's house. Behind the door was a magical world filled

--- TEXT 5 ---
Original Text:
 Customer: I am facing issues with my internet connection.
Support: Please restart your router and check the cables.
Customer: I tried that but it still does not work.
Support: I will register a complaint and a technician will visit your location tomorrow.

Summary:
 Customer: I am facing issues with my internet connection. Support: Please restart your router and check the cables. Customer: I
```

```
# Test different summary lengths

sample_text = texts[0]

# Function to generate summary for different lengths
def generate_summary(text, min_len, max_len):
    inputs = tokenizer([text], max_length=1024, return_tensors="pt", truncation=True)
    summary_ids = model.generate(
        inputs["input_ids"],
        num_beams=4,
        min_length=min_len,
        max_length=max_len,
        early_stopping=True
    )
    return tokenizer.decode(summary_ids[0], skip_special_tokens=True)

short_summary_text = generate_summary(sample_text, min_len=10, max_len=30)
medium_summary_text = generate_summary(sample_text, min_len=20, max_len=60)
long_summary_text = generate_summary(sample_text, min_len=40, max_len=120)
```

```
print("\n🔹 SHORT SUMMARY:\n", short_summary_text)
print("\n🔹 MEDIUM SUMMARY:\n", medium_summary_text)
print("\n🔹 LONG SUMMARY:\n", long_summary_text)
```

```
🔹 SHORT SUMMARY:
 Artificial Intelligence is transforming industries by enabling machines to perform tasks that traditionally required human inte

🔹 MEDIUM SUMMARY:
 Artificial Intelligence is transforming industries by enabling machines to perform tasks that traditionally required human inte

🔹 LONG SUMMARY:
 Artificial Intelligence is transforming industries by enabling machines to perform tasks that traditionally required human inte
```

```
long_text = """Artificial Intelligence (AI) has rapidly evolved over the past few decades...
AI is used in healthcare for diagnosis, in finance for fraud detection, and in education
for personalized learning. However, challenges such as bias, lack of transparency, and
ethical concerns must be addressed. Governments and researchers are working to ensure
responsible AI development. Additionally, advancements in machine learning and deep learning
have significantly improved AI capabilities, making it more efficient and accurate.
The future of AI includes automation, improved decision-making systems, and integration
into daily life. Despite its benefits, AI must be carefully regulated to prevent misuse.""" * 3  # make it long

summary_long_text = generate_summary(long_text, min_len=50, max_len=150)

print("\nLONG TEXT SUMMARY:\n", summary_long_text)
```

```
LONG TEXT SUMMARY:
 Artificial Intelligence (AI) has rapidly evolved over the past few decades. It is used in healthcare for diagnosis, in finance
```

```
# Save summaries to file

with open("summaries.txt", "w") as f:
    for i, summary in enumerate(results):
        f.write(f"Text {i+1} Summary:\n{summary}\n\n")

print("Summaries saved successfully!")
```

```
Summaries saved successfully!
```